

Still, the formulation of global hyperbolicity would be rather indirect. We would prefer a formulation phrased directly in terms of continuity properties of  $d$ .

To start with, we shall explore some causality properties expressed via the Lorentzian distance function.

**Proposition 2.1.** *On a spacetime  $(M, g)$  for any two events  $p, q \in M$ , the inclusion  $I^+(q) \subset I^+(p)$  is equivalent to the property  $d_q \leq d_p$ . Dually, the inclusion  $I^-(p) \subset I^-(q)$  is equivalent to the property  $d^p \leq d^q$ .*

This result is a bit surprising since the function  $d$  contains much more information than the chronological relation. Note that we do not assume that  $d$  is finite or continuous.

*Proof.* Suppose that  $I^+(q) \subset I^+(p)$  and let  $r \in M$ . If  $r \notin I^+(q)$  then  $d_q(r) = 0$  and the inequality we wish to prove follows. If  $r \in I^+(q)$  for any  $m > 0$ ,  $m < d_q(r)$ , there is a timelike curve  $\gamma : [0, 1] \rightarrow M$  connecting  $q = \gamma(0)$  to  $r = \gamma(1)$  of Lorentzian length  $\ell(\gamma) > m$ . Let  $\delta > 0$  be such that  $\ell(\gamma|_{[\delta, 1]}) > m$ . But  $\gamma(\delta) \gg q$  hence  $\gamma(\delta) \gg p$ , which implies  $d_p(r) \geq \ell(\gamma|_{[\delta, 1]}) > m$ . Since

$$m < d_q(r)$$

is arbitrary,  $d_p(r) \geq d_q(r)$ . By the arbitrariness of  $r$ ,  $d_q \leq d_p$ .

For the converse, suppose that  $d_q \leq d_p$ , then if  $r \in I^+(q)$  we have  $d_q(r) > 0$  so  $d_p(r) > 0$  which implies  $r \in I^+(p)$ . By the arbitrariness of  $r$ ,  $I^+(q) \subset I^+(p)$ .  $\square$

We recall [43, Def. 4.46] that a spacetime is *future distinguishing* if  $I^+(p) = I^+(q) \Rightarrow p = q$  (and similarly in the past case). A spacetime is *weakly distinguishing* if  $I^+(p) = I^+(q)$  and  $I^-(p) = I^-(q) \Rightarrow p = q$ , cf. [43, Def. 4.47]. A spacetime is *distinguishing* if  $I^+(p) = I^+(q)$  or  $I^-(p) = I^-(q) \Rightarrow p = q$ , cf. [43, Def. 4.59].

It is convenient to introduce the following notions

**Definition 2.2.** We say that the *Lorentzian distance future distinguishes events* (the spacetime is *future  $d$ -distinguishing*) if  $d_p = d_q \Rightarrow p = q$ , and that is *past distinguishes events* if  $d^p = d^q \Rightarrow p = q$ . The spacetime is  *$d$ -distinguishing* if it is both future and past  $d$ -distinguishing, namely:  $d^p = d^q$  or  $d_p = d_q \Rightarrow p = q$ . We say that the spacetime is *weakly  $d$ -distinguishing* if  $d^p = d^q$  and  $d_p = d_q \Rightarrow p = q$ .

Note that  $d$ -distinction implies ‘future or past  $d$ -distinction’ which implies weak  $d$ -distinction.

*Remark 2.3.* Weak  $d$ -distinction is called just  $d$ -distinction (or it is said that the Lorentzian distance distinguishes events) in [45, 13] as this is the only version that plays a role in those works.

An immediate consequence of Prop. 2.1 is

**Corollary 2.4.** *On a spacetime  $(M, g)$  (future/past/weak)  $d$ -distinction is equivalent to (resp. future/past/weak) distinction.*